

Ivan Shamrai

Senior/Lead SDET | Test Automation | Python & Go | Distributed Systems

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PROFILE

Senior/Lead SDET with 15+ years in software engineering, infrastructure, and QA automation, including 12+ years in test engineering. Builds high-throughput test platforms, performance and resiliency pipelines, and developer tooling in Python and Go for trading, banking, embedded, and distributed systems. Combines hands-on engineering with 4+ years of QA leadership, including teams of 5-10 engineers.

300K messages/sec load generation

Message preparation optimized from 146 us to 6 us

500 GB production trading-data analysis

CORE COMPETENCIES

Languages	Python, Go, Bash/Shell, SQL/PLSQL	Test Engineering	pytest, JMeter, k6, API and integration testing, load and performance testing, resiliency testing
Trading Systems	FIX, ITCH, exchange-native protocols, market data, post-trade and clearing workflows	Cloud and Delivery	Kubernetes, Docker, AWS (EKS, Lambda, EC2), GitLab CI, Jenkins, Azure DevOps
Data and Observability	PostgreSQL, MySQL, MSSQL, ClickHouse, InfluxDB, Grafana, NATS, ZeroMQ		

PROFESSIONAL EXPERIENCE

Software Development Engineer in Test Exactpro Systems LLC

May 2024 - Present

Develops and operates performance, load, resiliency, and protocol-testing tooling for exchange and post-trade platforms.

Exchange Load Injection Platform

- Developed and supported a Go load-injection tool for an exchange platform, with distributed-system monitoring and performance tuning.
- Sustained throughput up to 300K messages/sec.
 - Analyzed 500 GB of raw production trading PCAP data to build realistic load profiles.
 - Reduced message preparation latency from 146 us to 6 us through parallelization and optimization.

Go, PCAP, distributed monitoring, performance engineering

Historical Trading Data Replay

- Built tooling to replay real historical trading data and validate new releases against production-like market activity.
- Replays a full trading day in 30 minutes to 2 hours, depending on volume.
 - Supports all instruments or targeted instrument subsets for focused validation.

Go, trading data, release validation, load testing

Kubernetes Multi-Matching-Engine Test Automation

- Automated configuration, reference-data generation, load generation, performance-metric calculation, and order execution/market-data reports.
- Supports realistic deployments with up to four matching engines.
 - Reduced manually maintained test variants from 30 to 10.

Kubernetes, Python, Go, performance analytics, test data generation

Post-Trade Platform Testing

- Automates performance and resiliency validation for post-trade and clearing workflows across configurable load pipelines.
- Measures p50, p99, average, and maximum latency using k6 and FIX test flows.
 - Runs fully automated long-duration tests for up to 12 hours.
 - Validates AWS infrastructure spanning EKS, Lambda, databases, and EC2.

k6, FIX, AWS, EKS, Lambda, EC2, resiliency testing

EARLIER PROFESSIONAL EXPERIENCE

Software Development Engineer in Test / DevOps

Jun 2022 - Apr 2024

Kaspersky

- Built a Python/pytest automation framework and maintained hardware-in-the-loop test benches for automotive and embedded systems.
- Created a development-board CLI that cut setup from 15-20 minutes to 2-3 minutes and parallel test execution from 2 hours to 15 minutes.
- Integrated Vault-managed secrets and optimized CI images, caching, and parallelization to reduce build time by 30%.

Python, Go, pytest, Azure DevOps, Vault, Docker, Grafana, InfluxDB, CAN, UDS, MQTT

Senior QA Automation Engineer

Sep 2021 - Jun 2022

Lenovo

- Stabilized a Python/pytest API and integration framework, raising automated pass rate to about 95% and cutting execution time by 25-35%.
- Migrated CI/CD from Bamboo to Jenkins and reverse-engineered a ZeroMQ protocol to expand internal-service coverage by about 30%.

Python, pytest, REST API, ZeroMQ, Jenkins, Bamboo, JWT, JSON Schema

Quality Assurance Team Lead

Aug 2020 - Sep 2021

BPC Banking Technologies

- Led QA process and delivery improvements, increasing team throughput by about 15-20% through workload and cycle-time analysis.
- Automated isolated Kubernetes test environments and branch-level CI checks, reducing wasted QA effort on broken builds by 30-40%.

Python, Go, Kubernetes, GitLab CI, Jenkins, Jira, Confluence, PLSQL

QA Specialist to QA Team Lead

Feb 2014 - Aug 2020

Exactpro

- Progressed through three roles while testing low-latency exchange systems and automating FIX, ITCH, and native-protocol workflows.
- Built load, analysis, environment, and monitoring tools; automated non-functional testing across four projects.
- Led and mentored QA engineers while introducing InfluxDB/Grafana visibility and reducing manual environment support by 2-4x.

Python, Go, Bash, JMeter, FIX, ITCH, Ansible, InfluxDB, Grafana, SQL

Earlier Infrastructure Experience

Dec 2010 - Feb 2014

Home Credit & Finance Bank; RIHMI-WDC

- Supported banking middleware and research infrastructure across Linux/Unix, WebLogic/OSB, VMware, networks, databases, monitoring, and automation.

RHEL, FreeBSD, WebLogic, Oracle Service Bus, VMware ESXi, Python, Jython, Zabbix

SELECTED PROJECTS

mdify

Document-to-Markdown CLI

Built and maintains a cross-platform Python CLI that converts documents and images to Markdown through containerized processing, runtime auto-detection, broad format support, and automated tests.

Python, CLI design, Docker, Podman, document processing, pytest

mnemofy

Local Speech and Meeting-Notes Pipeline

Built a Python CLI for local speech transcription and structured meeting-note generation with timestamps, topics, decisions, action items, and optional local or API-based language-model processing.

Python, faster-whisper, ffmpeg, Ollama, OpenAI API, structured text generation

Aequitas

Event-Driven Market-Neutral Trading Platform

Designed a Go-based platform with separated market-data, strategy, and execution services, event streaming, time-series storage, persistent state, and risk-first position lifecycle controls.

Go, NATS, ClickHouse, PostgreSQL, Docker, WebSocket market data

INDEPENDENT RESEARCH

Evaluation-First Retry Policy Evolution

Designed a reproducible study of automated software evolution using a narrow mutation surface, deterministic train/holdout evaluation, hard safety gates, and systematic analysis of overfitting, reward hacking, evaluator weaknesses, and model failure modes.

Python, pytest, OpenEvolve, local language models, reliability engineering

EDUCATION

Master of Science, Information Systems and Technologies

Obninsk Institute for Nuclear Power Engineering 2010 - 2012

Bachelor of Science, Information Systems and Technologies

Obninsk Institute for Nuclear Power Engineering 2006 - 2010

Languages: English - working proficiency; Russian - native

Interests: AI/LLM systems; Embedded systems